

Weather Conditions and Climate Change with ClimateWins

Priya Agrawal

July 15, 2025

A satellite image of Europe and surrounding regions, including North Africa and the Middle East, showing landmasses, oceans, and cloud patterns. The image is positioned on the left side of the slide, partially obscured by a green geometric design.

Introduction

- ▶ Climate change has been a great concern for many years and continues to be all over the world
- ▶ Over the past 10-20 years, there has been an increase in extreme weather events due to climate change, which is especially of concern to ClimateWins
- ▶ ClimateWins aims to use machine learning to assist with predicting the consequences of climate change in Europe

Hypotheses



If temperatures increase every year, then extreme weather events will also increase.



Machine learning algorithms will accurately predict if the weather on a day is pleasant or unpleasant.



The accuracy of weather predictions will differ for the European regions because of differences in climate.

Data

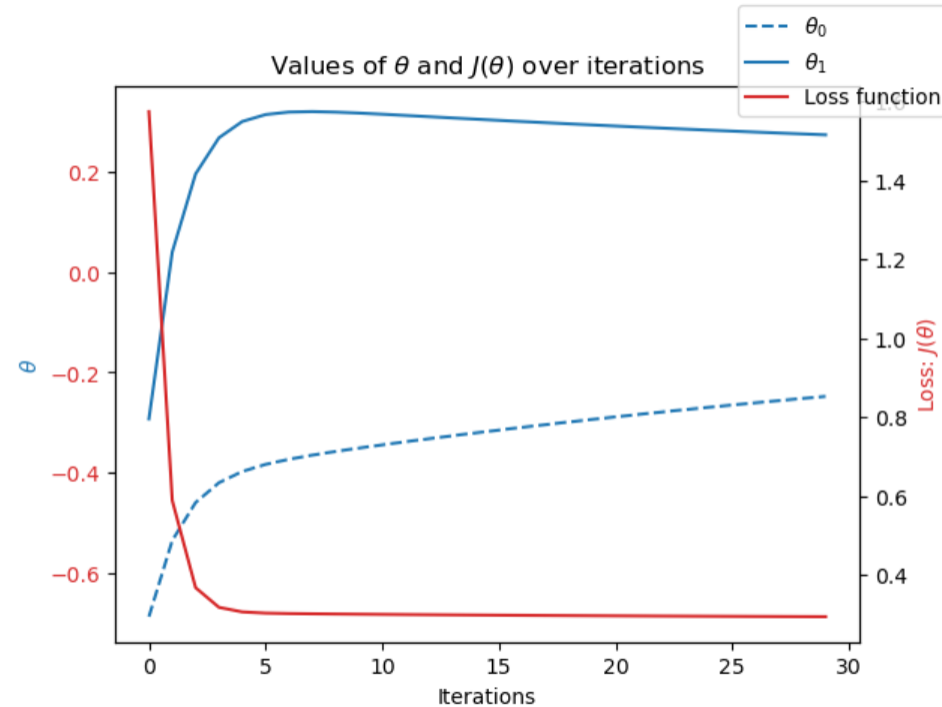
- ▶ Temperature data set:
 - ▶ Data was collected by the European Climate Change Assessment & Data Set project
 - ▶ Includes weather observations from 18 weather stations located in Europe
 - ▶ Data ranges from the late 1800s to 2022
 - ▶ Includes variables such as global radiation, wind speed, temperature and more

Data Bias and Accuracy

- ▶ **Collection bias:** Collection methods for older data may be unreliable and more prone to human error as collection was likely done manually, resulting in potential inaccuracies in older data
- ▶ **Temporal bias:** Older data in the data set, especially data that is more than 50 years old, may not accurately represent current weather conditions, which can lead to inaccurate weather predictions
- ▶ **Selection bias:** Larger and more developed areas in Europe may be overrepresented and smaller and less developed areas may be underrepresented due to the location of the weather stations, leading to inaccurate weather predictions in areas that are smaller and less developed

Data Optimization

- Data optimization was done using Gradient Descent
- Many changes were made to the parameters to minimize the loss function
- The iterations to the parameters brought the loss function close to 0



Algorithms

- ▶ Three supervised machine learning algorithms were used: K-Nearest Neighbour, Decision Tree, and Artificial Neural Network
- ▶ These algorithms are more suitable for weather prediction in comparison to other supervised machine learning algorithms since:
 - ▶ **K-Nearest Neighbour:** Does not require much training as it uses patterns found in the data to generate predictions, does not make any assumptions about the data distribution (non-parametric), and works well with spatial and temporal data.
 - ▶ **Decision Tree:** Is robust to outliers, easy to interpret because of the structure, and can easily handle mixed-type data and non-linear relationships.
 - ▶ **Artificial Neural Network:** Finds complex, non-linear relationships between variables, identifies relationships between predictor variables, and becomes better as it works with greater amounts of data.

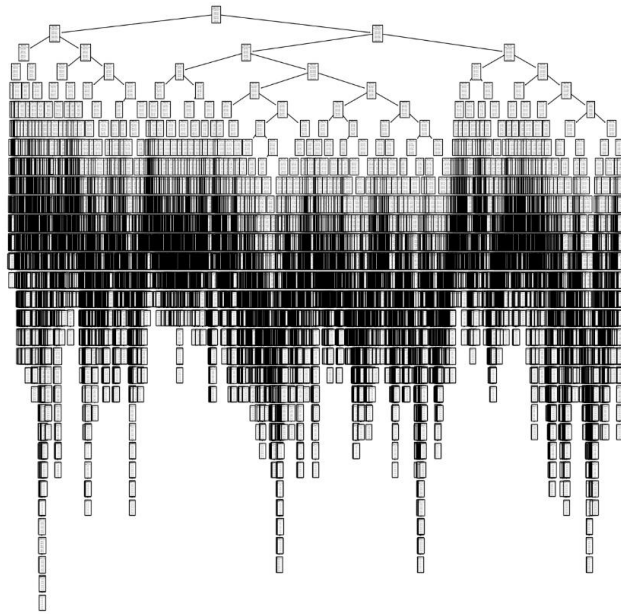
Algorithms

- ▶ Each of the K-Nearest Neighbour, Decision Tree and Artificial Neural Network algorithms were tested to see which one would best predict days with pleasant and unpleasant weather in the different regions of Europe
- ▶ Tested each algorithm to see which one had the best success measure
 - ▶ Success measure used is the accuracy score which is a percentage out of 100
 - ▶ The algorithm with the highest accuracy score is the most effective
 - ▶ The algorithm with the highest accuracy score is the one that ClimateWins should use to predict the consequences of climate change

K-Nearest Neighbour

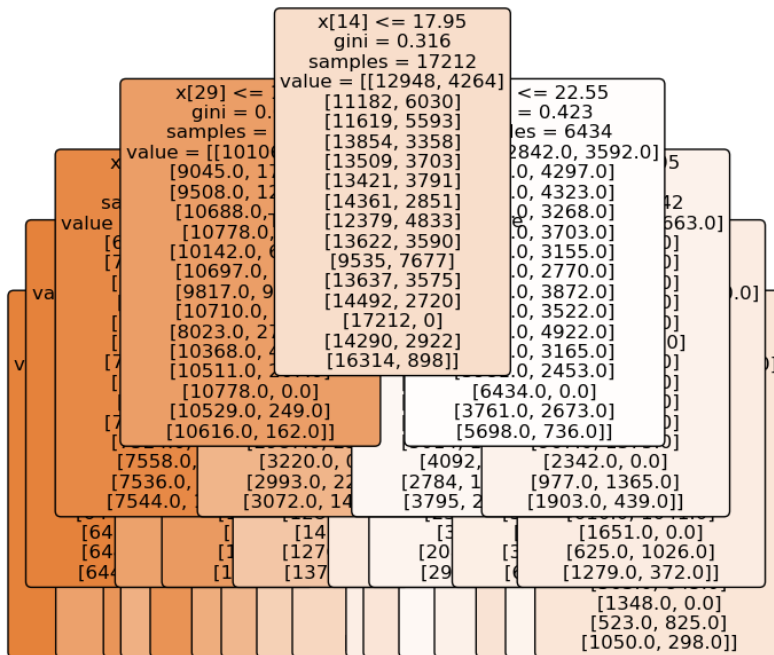
Weather Station	Accurate Predictions		False Positives	False Negatives	Accuracy Rate
Basel	3917	961	421	439	85%
Belgrade	3252	1544	524	418	84%
Budapest	3424	1462	476	376	85%
<u>Debilt</u>	4320	723	317	378	88%
Dusseldorf	4164	810	343	421	87%
Heathrow	4138	744	432	424	85%
Kassel	4563	614	252	309	90%
Ljubljana	3740	1180	455	363	86%
Maastricht	4253	824	309	352	88%
Madrid	2750	2261	418	309	87%
<u>Munchenb</u>	4237	792	309	400	88%
Oslo	4637	512	242	347	90%
<u>Sonnblick</u>	5738	0	0	0	100%
Stockholm	4483	607	283	365	89%
Valentia	5404	74	58	202	96%
				Average	89%

- ▶ The K-Nearest Neighbour algorithm uses a distance formula to determine the distance between new data points and other data points. A data point is then placed in a group with data points that are nearby depending on the neighbour count of each group.
- ▶ This algorithm was used to predict the number of days with pleasant and unpleasant weather at 15 weather stations in Europe
- ▶ The average accuracy of the algorithm is 89%
- ▶ Sonnblick's 100% accuracy rate indicates that overfitting is likely happening



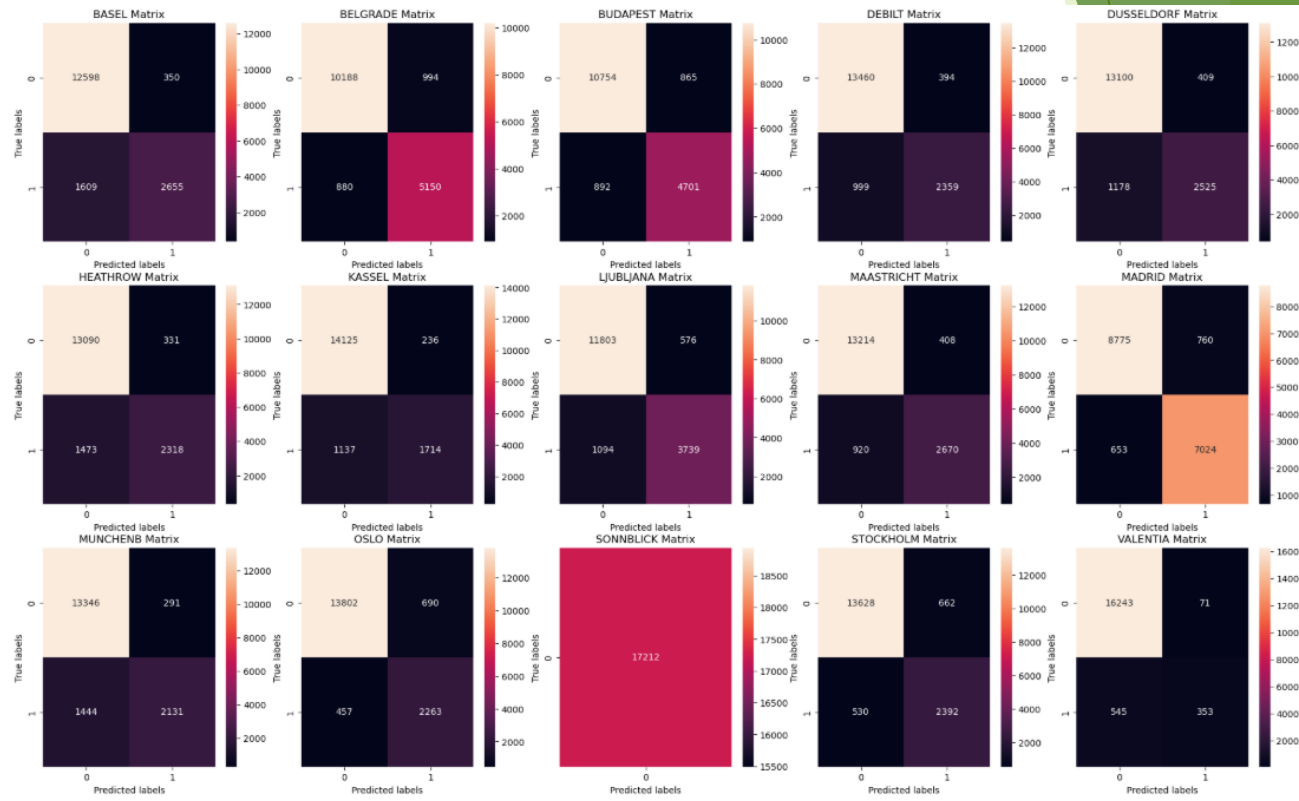
Decision Tree

- ▶ The Decision Tree algorithm makes predictions by starting with a root question that branches off into more questions about the data that become increasingly specific. Data points flow into different branches of the tree based on their features until they reach a leaf which is a stopping point.
- ▶ This algorithm was used to predict the number of days with pleasant and unpleasant weather at 15 weather stations in Europe
- ▶ The accuracy of the training data is 46% and the testing data is 47%
- ▶ This Decision Tree is quite intricate as it has too many branches and leaves and needs to be pruned



The first 4 rows of the decision tree.

Artificial Neural Network



For each confusion matrix, the upper left box represents accurate predictions for unpleasant weather, the upper right box represents inaccurate predictions of pleasant weather, the lower left box represents inaccurate predictions of unpleasant weather, and the lower right box represents accurate predictions for pleasant weather.

- ▶ The Artificial Neural Network algorithm multiplies inputs (received in the input layer) by weights into a hidden layer. Data is processed in the hidden layer, and the results are sent to the output layer which produces the prediction.
- ▶ This algorithm was used to predict the number of days with pleasant and unpleasant weather at 15 weather stations in Europe
- ▶ The best accuracy scores obtained from a few tests with different parameters is 52% for the training data and 49% for the testing data
- ▶ The confusion matrices display accurate predictions, false positives, and false negatives

Most Effective Algorithm

- ▶ The K-Nearest Neighbour algorithm has an accuracy rate of 89% which is the highest of all three algorithms, meaning that it is the most effective algorithm
- ▶ Accuracy rate for K-Nearest Neighbour is significantly higher than the accuracy rates of the Decision Tree and Artificial Neural Network algorithms
- ▶ ClimateWins should use the K-Nearest Neighbour algorithm to make weather predictions in Europe

Summary

Hypotheses

- ▶ If temperatures increase every year, then extreme weather events will also increase.
- ▶ Machine learning algorithms will accurately predict if the weather on a day is pleasant or unpleasant.
- ▶ The accuracy of weather predictions will differ for the European regions because of differences in climate.

Methods

- ▶ K-Nearest Neighbour, Decision Tree, and Artificial Neural Network algorithms were tested to see which is the most effective
- ▶ K-Nearest Neighbour is the most effective algorithm as it has the highest accuracy

Next Steps

- ▶ Collect more varied and diverse data to train the algorithm on including data from additional weather stations
- ▶ Utilize unsupervised machine learning algorithms
- ▶ Include other variables in the analysis
- ▶ Predict weather at each of the weather stations



Thank you!

Please contact me at priya.agrawal0929@gmail.com if you have any questions.

Take a look at my [GitHub](#) for the Python code used in this project.